JGi JAIN DEEMED-TO-BE UNIVERSITY

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

TEAM - 7

POLYGLOT-TALK

AN AI INTERPRETATION MODEL

GUIDED BY,

Dr. M. Sayeekumar
Associate Professor, CSE - AIML

PRESENTED BY,

Dhritiraj Lahakar (23BTRCL041)
Panshul P Sarma (23BTRCL063)
Shrijith S Menon (23BTRCL022)
Soumyajit Chanda (23BTRCL117)
Yash Karecha (23BTRCL088)

OVERVIEW

An overview of the topic covered in this presentation.

ABSTRACT

INTRODUCTION

SYSTEM REQUIREMENT

LITERARY REVIEW

EXISTING SYSTEMS

BLOCK DIAGRAM

PROPOSED SYSTEM

CONCLUSION

REFERENCES

POLYGLOT-TALK

ABSTRACT

In a world that is becoming increasingly interconnected, real-time multilingual communication remains a significant challenge.

This project aims to develop a mobile application for real-time speech-to-speech (S2S) interpretation, enabling seamless conversations between speakers of different languages.

The application is designed to function as a calling app, allowing users to communicate naturally without delays.

The app employs a buffer-based processing approach to achieve low-latency translation, where speech is captured in small intervals (e.g., 2-3 seconds). Each buffer is processed in parallel using multi-threading, ensuring that transcription, translation, and text-to-speech (TTS) synthesis occur simultaneously, minimizing lag. The system dynamically switches between on-device processing and cloud-based processing depending on network conditions, prioritizing local execution for speed and privacy.

The project involves training and optimizing automatic speech recognition (ASR) models, machine translation (MT) models, and TTS models for real-time inference. It leverages efficient deep learning architectures, ensuring compatibility with low-end mobile devices while maintaining high accuracy. Additionally, the app will feature an intuitive user interface, making it accessible for various real-world applications, including international business, travel, and accessibility for non-native speakers.

INTRODUCTION

- With globalization accelerating, the need for instant multilingual communication is greater than ever. Current speech translation systems primarily rely on sentence-based transcription, where speech is processed in complete sentences before translation and playback. While effective for some applications, this method introduces latency issues and disrupts the natural flow of conversations.
  In real-time phone calls, such delays become impractical, making seamless and low-latency speech translation a critical requirement.

- This project proposes a real-time speech-to-speech translation app, designed specifically for live phone conversations. The system uses a buffer-based streaming approach, where speech is captured in short overlapping buffers instead of waiting for full sentences. Each buffer is processed through a multi-threaded pipeline, where speech-to-text conversion (ASR), translation (MT), and speech synthesis (TTS) occur in parallel, significantly reducing processing time. Additionally, context continuity is maintained across buffers to ensure natural and fluent translations.

SYSTEM REQUIREMENTS

HARDWARE REQUIREMENTS

- Processor: Quad-core CPU or higher
- RAM: Minimum 4GB (8GB recommended)
- Storage: 500MB of free space
- Microphone & Speaker
- Internet Connectivity (Optional)

- For Cloud Processing (if used)
  - CPU: 16-core
  - RAM: 16GB or more
  - GPU: NVIDIA Tesla T4 or higher
  - Network: Low-latency infrastructure

SOFTWARE REQUIREMENTS

- OS: Android 8.0+ / iOS 12+
- Libraries: Whisper, Argos, VITS
- Backend: FastAPI / Flask

- Cloud Backend
  - ASR: Whisper (Faster-whisper)
  - Translation: Argos
  - TTS: Facebook / mms-tts (VITS)
  - Database: Firebase Firestore

LITERARY REVIEW

- Guan, Yong & Zheng, Lin & Tian, Jilei. (2010). Real-time speaker adapted speech to speech translation system in mobile environment. 577 - 580. 10.1109/ICOSP.2010.5656047.
- Yang, N. Optimization and Performance Analysis of Real-Time Speech Translation Systems Based on Mobile Technology. Int. J. Interact. Mob. Technol. 2024, 18, pp. 57-71.
- S, Karunya & M, Jalakandeshwaran & Babu, Thanuja & R, Uma. (2023). AI-Powered Real-Time Speech-to-Speech Translation for Virtual Meetings Using Machine Learning Models. 1-6. 10.1109/ICCEBS58601.2023.10448600.
- R. Prasad, et al., “Real-Time Speech-to-Speech Translation for PDAs,” in IEEE International Conference on Portable Information Devices, 2007, pp. 1–5.

EXISTING TECHNOLOGIES

Language Barriers in Communication: Real-time conversations between speakers of different languages are difficult due to the lack of instant translation.
High Latency in Existing Solutions: Most translation tools require users to pause after speaking, disrupting the flow of conversation.

Independent Chunk Processing: Traditional translation models process speech in isolated chunks, leading to loss of context and unnatural translations.

Internet Dependency: Cloud-based speech translation solutions introduce latency and inconsistency due to network dependence, making them unreliable in areas with poor connectivity.

Sentence-Based Transcription Limitation: Existing speech translation systems often rely on full-sentence transcription before translating, leading to delays and unnatural pauses. Our system uses a buffer-based approach for continuous, low-latency processing without waiting for sentence completion.

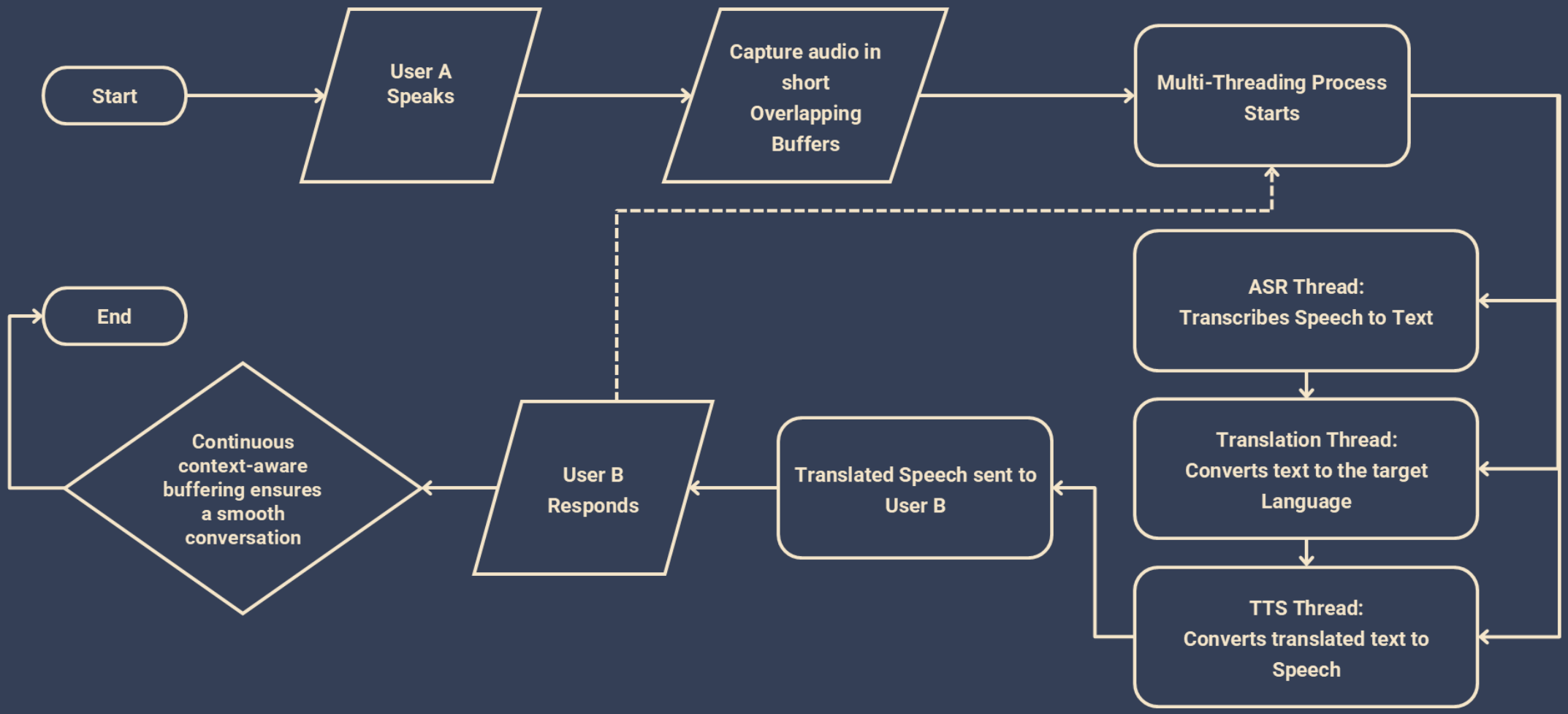
POLYGLOT-TALK

PROPOSED SYSTEM

We propose a real-time speech-to-speech translation calling app designed to enable seamless, live multilingual conversations.
The system ensures smooth interactions by incorporating:

- **Buffer-Based Streaming**:
  - The app captures speech in short overlapping buffers instead of waiting for full sentences, ensuring continuous processing without unnatural pauses.
- **Parallel Processing**:
  - Audio transcription (ASR), translation, and speech synthesis (TTS) occur simultaneously in separate threads to reduce delay.
- **Context Awareness**:
  - Each buffer is processed with dependency on previous buffers, ensuring that translations maintain linguistic coherence.
- **Hybrid Processing Mode**:
  - Local Processing: For speed and offline availability, speech processing happens on-device when possible.
  - Cloud Support: If more computational power is needed or for improved accuracy, the system can switch to cloud-based processing dynamically.
- **Bidirectional Translation for Calls**:
  - Unlike standalone translation apps, this system is designed for real-time phone calls, ensuring that translated speech is sent directly to the other caller instead of playing it back to the speaker.

POLYGLOT-TALK



CONCLUSION

This project aims to create a real-time speech-to-speech translation calling app that enables seamless, natural conversations between speakers of different languages. By leveraging multi-threaded parallel processing, context-aware translation, and hybrid processing, the app will provide low-latency, high-quality translations across mobile and cloud platforms.

THANK YOU